

Iterated Conformal Infinity Is Not Iterated Holography

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Abstract

We introduce *depth-two conformal infinity*: a conformally compact Einstein manifold whose conformal boundary admits a complete asymptotically hyperbolic representative that has a conformal infinity of its own. Such structures lead naturally to cornered asymptotically hyperbolic geometry. The resulting hierarchy, however, does not by itself define a recursive holographic correspondence. The second asymptotic geometry needs a distinguished representative of the first boundary’s conformal class, whereas ordinary conformal infinity records only the class. In the maximally symmetric model, $\text{AdS}_d \subset \text{AdS}_{d+1}$ carries its own conformal infinity as a codimension-one locus of the parent boundary. That incidence structure points toward a relative or defect sector, not a second, autonomous duality.

Introduction

What happens when the boundary of a spacetime has a boundary of its own?

Holography tempts us with a tidy answer. If a bulk can be encoded on its boundary [1, 2, 3], perhaps that boundary can itself serve as a bulk, and its boundary in turn as another:

$$X^{d+1} \longrightarrow M^d \longrightarrow N^{d-1} \longrightarrow \dots$$

The picture has real charm. Geometry becomes a stack of descriptions, each boundary concealing a further interior behind it, in keeping with the founding spirit of holography.

Geometry is less obliging than the metaphor. Conformal infinity yields a conformal class, not a metric, and a class does not say which of its representatives ought to become the next bulk. The recursion therefore meets a precise obstruction long before any question about a second CFT can be posed. This note isolates that obstruction and asks what it means.

1. Depth-two conformal infinity

Let (X^{d+1}, g_+) be a conformally compact Einstein manifold [9, 5],

$$\text{Ric}(g_+) = -\frac{d}{L^2}g_+. \quad (1)$$

Thus X is the interior of a compact manifold $\overline{X} = X \cup M$, and for a defining function x of M ,

$$g_+ = \frac{L^2}{x^2} \overline{g}, \quad [\gamma] = [\overline{g}|_{TM}]. \quad (2)$$

The conformal infinity is the conformal manifold $(M^d, [\gamma])$. The boundary datum is the *class* $[\gamma]$, not a distinguished metric γ , and everything below turns on that distinction.

We now ask whether the boundary can carry asymptotic structure of its own.

Definition 1 (Depth-two conformal infinity). *The Poincaré–Einstein space (X^{d+1}, g_+) has a depth-two conformal infinity if its conformal infinity $(M, [\gamma])$ contains a complete representative $\gamma_+ \in [\gamma]$ admitting a compactification*

$$\gamma_+ = \frac{\ell^2}{y^2} \overline{\gamma}, \quad \overline{M} = M \cup N, \quad \overline{\gamma}|_{TN} \in [h], \quad (3)$$

where y is a defining function for $N = \partial \overline{M}$. If, in addition,

$$\text{Ric}(\gamma_+) = -\frac{d-1}{\ell^2} \gamma_+, \quad (4)$$

we call the second level Poincaré–Einstein.

The resulting structure is

$$(X^{d+1}, g_+) \longrightarrow (M^d, [\gamma], \gamma_+) \longrightarrow (N^{d-1}, [h]). \quad (5)$$

The middle metric γ_+ is no ornament. It records the extra choice that makes the second arrow meaningful: ordinary conformal compactification delivers $[\gamma]$ and is silent on which $\gamma_+ \in [\gamma]$ should serve as a complete asymptotically hyperbolic metric. The recursive question therefore differs from a mere repetition of the first compactification:

$$[\gamma] \quad \neq \quad \gamma_+.$$

2. Corners and two asymptotic scales

A simultaneous compactification is naturally described by two boundary-defining functions,

$$x \geq 0, \quad y \geq 0.$$

The compactified geometry then has boundary faces

$$F_x = \{x = 0\}, \quad F_y = \{y = 0\}, \quad C = F_x \cap F_y, \quad (6)$$

with

$$\dim X = d+1, \quad \dim F_x = d, \quad \dim C = d-1. \quad (7)$$

The codimension-two locus C is where the two asymptotic descriptions meet.

Schematically, the desired local structure is

$$g_+ \sim \frac{L^2}{x^2} \left[dx^2 + \frac{\ell^2}{y^2} (dy^2 + h_y) \right], \quad h_y = h_{(0)} + O(y). \quad (8)$$

This is a nested asymptotic ansatz and nothing more; we do not claim that an arbitrary h_y solves the Einstein equation. Its role is modest: it displays the two defining functions and the corner conditions that a genuine solution would have to meet.

The relevant local geometry already belongs to the theory of cornered asymptotically hyperbolic manifolds. McKeown constructed normal forms adapted to a finite boundary meeting an asymptotic boundary and developed formal Einstein expansions at the resulting corner [7, 8], and smooth compactification there is subject to nontrivial obstructions. Our question is narrower and, we think, sharper: what further meaning can be assigned when the first boundary face is itself required to carry a complete asymptotically hyperbolic geometry?

3. The representative-selection obstruction

The obstruction is not that depth-two geometry fails to exist. Whether a conformal class contains a complete Poincaré–Einstein representative is a legitimate geometric question. The difficulty is that ordinary conformal infinity does not select such a representative canonically.

Changing the defining function in (2) changes the boundary representative according to

$$\gamma \mapsto \hat{\gamma} = e^{2\omega} \gamma. \quad (9)$$

For $d > 2$ the Ricci tensor transforms as [9]

$$\begin{aligned} \text{Ric}(\hat{\gamma}) &= \text{Ric}(\gamma) - (d-2)\nabla^2\omega - \gamma\Delta\omega \\ &\quad + (d-2)(d\omega \otimes d\omega - |\nabla\omega|^2\gamma). \end{aligned} \quad (10)$$

Hence even when γ is Einstein, a generic Weyl-related representative is not. If both γ and $\hat{\gamma}$ are Einstein, the trace-free part of (10) forces

$$(\nabla^2\omega - d\omega \otimes d\omega)_{\text{TF}} = 0. \quad (11)$$

Equivalently, with $u = e^{-\omega}$ one has $\nabla^2 u = e^{-\omega}(d\omega \otimes d\omega - \nabla^2\omega)$, so (11) says $(\nabla^2 u)_{\text{TF}} = 0$, the classical condition for two Einstein metrics to be conformally related [6]. It is a strong overdetermined system. The Weyl freedom of a conformal class is vastly larger than the freedom that preserves an Einstein representative, and a second Einstein structure therefore calls for a distinguished representative rather than following from Weyl equivalence alone.

This yields the central statement.

Proposition 1 (Representative-selection obstruction). *Let (X^{d+1}, g_+) have conformal infinity $(M, [\gamma])$. Suppose $[\gamma]$ contains a complete Poincaré–Einstein representative γ_+ with conformal infinity $(N, [h])$. Then the resulting depth-two structure contains additional representative data: it is not invariant under arbitrary changes of representative in $[\gamma]$. Consequently, the conformal-boundary data $(M, [\gamma])$ alone do not canonically determine the second Poincaré–Einstein geometry.*

The proposition does not deny that $[\gamma]$ might contain a preferred representative under some further geometric prescription. It says that any such prescription is extra structure.

A recursive holographic reading would thus need a map

$$[\gamma] \xrightarrow{\mathcal{S}} \gamma_+, \quad (12)$$

where $\mathcal{S}$ selects a complete AH or PE representative. Ordinary conformal-boundary theory supplies none. Such a selection map is the *sine qua non* of any iterated dictionary, and it is exactly what the hierarchy (5) lacks.

At this point the elegant recursion

$$X \longrightarrow M \longrightarrow N$$

stops being automatic. The second step is not a second copy of the first; it is a new geometric problem.

4. The maximally symmetric test

Pure AdS makes the distinction unusually transparent. In the embedding description,

$$\text{AdS}_d = \text{AdS}_{d+1} \cap \{X_{d+1} = 0\}, \quad (13)$$

where the intersection is totally geodesic. Equivalently,

$$ds_{\text{AdS}_{d+1}}^2 = L^2 [d\rho^2 + \cosh^2 \rho \, ds_{\text{AdS}_d, \text{unit}}^2]. \quad (14)$$

A fixed- ρ slice is intrinsically AdS_d , with radius

$$\ell_d(\rho) = L \cosh \rho.$$

This looks, at first glance, like exactly the recursive geometry one hoped for. The catch is that the slice is an interior hypersurface, and its conformal infinity fills only a codimension-one locus of the ambient conformal boundary,

$$\partial_\infty \text{AdS}_d \subset \partial_\infty \text{AdS}_{d+1}. \quad (15)$$

The lower-dimensional infinity is thus a subgeometry of the first boundary, not a second, independent conformal boundary. It is also worth seeing how the model realises the definition: as $\rho \rightarrow \pm\infty$ the metric (14) shows that each open half of the parent boundary is conformal to unit AdS_d , so one may take $\gamma_+ = \text{AdS}_d$ and N the interface between the two halves. The selection of γ_+ is supplied here by the slicing itself, that is, by the symmetry.

The maximally symmetric model thereby separates two ideas:

an AdS subgeometry whose boundary is another boundary
 $\neq$
the entire boundary becoming a new AdS bulk.

The first arises naturally; the second demands extra structure.

This incidence picture is suggestive of boundary, defect, or relative degrees of freedom, and it resembles the holographic constructions for theories with boundaries or interfaces [13, 11, 12]. It should not be identified with them, however: those constructions introduce additional physical data specifying the boundary or interface.

5. What holography does, and does not, recurse

The semiclassical holographic dictionary [2, 3] can be written schematically as

$$Z_{\text{grav}}[X; \gamma] = Z_{\text{CFT}_d}[M; \gamma]. \quad (16)$$

A recursive interpretation would require a second dictionary,

$$Z_{\text{CFT}_d}[M; \gamma] \stackrel{?}{=} Z_{\text{grav}}[M; \gamma_+] \stackrel{?}{=} Z_{\text{CFT}_{d-1}}[N; h]. \quad (17)$$

But the middle term presupposes a distinguished representative

$$\gamma_+ = \mathcal{S}([\gamma]) \quad (18)$$

with the requisite complete asymptotic geometry. That selection is additional data, not something the conformal class $[\gamma]$ provides. The existence of a second conformal infinity therefore does not, of itself, define a second holographic dictionary:

depth-two conformal infinity $\nRightarrow$ iterated holography. (19)

To infer the right-hand side from the left is a *non sequitur*.

The point is more than technical. We usually picture holography as a descent in dimension, bulk information encoded on a boundary, and it is tempting to let the descent repeat. Geometry is more restrained. A boundary may possess asymptotic structure of its own without becoming an independent gravitational universe. What recurses is the *geometry*; whether the dictionary recurses is a separate question.

6. What would make the recursion possible?

The failure described above need not end the idea. It does identify the missing ingredient.

One would need a natural principle

$$[\gamma] \xrightarrow{\mathcal{S}} \gamma_+ \quad (20)$$

canonical enough to survive the conformal ambiguity. Candidate sources include a Yamabe–Einstein condition, extremisation of a renormalised geometric functional, a boundary stress-tensor condition, mixed boundary data, or a variational principle involving the corner.

A direct analytic test is available in a scalar field,

$$(\Delta_{g_+} - m^2)\Phi = 0. \quad (21)$$

Its first asymptotic expansion has the familiar form [2, 4]

$$\Phi \sim x^{d-\Delta}\phi_{(0)} + x^\Delta\phi_{(1)}, \quad m^2 L^2 = \Delta(\Delta - d). \quad (22)$$

If $\phi_{(0)}$ is in turn constrained by a wave equation on the asymptotically hyperbolic (M, γ_+) , it admits a second expansion,

$$\phi_{(0)} \sim y^{d-1-\delta}\psi_{(0)} + y^\delta\psi_{(1)}. \quad (23)$$

The composition of asymptotic data would then pose the question of whether a scattering or Dirichlet-to-Neumann operator on N can be made conformally natural [10].

That is a concrete problem for a longer paper. The present note claims less: the appealing recursive picture cannot be had for free.

7. Caveats and prior art

Several limitations deserve plain statement.

First, the proposition is deliberately modest. It shows non-invariance under Weyl rescaling, not non-existence and not non-uniqueness of a second Einstein representative. Whether a given class contains any complete Poincaré–Einstein representative at all is a strong requirement, as (11) indicates, and we prove no existence or classification result. In the symmetric model of Section 4 the choice is effectively forced by the symmetry group, which is why the example looks canonical; away from it, nothing comparable is guaranteed.

Second, the obstruction concerns the *bulk* reading. On the field theory side the partition function already depends on the representative through the Weyl anomaly, so the dependence of (16) on γ is not in itself the difficulty. What is new is that a gravitational interpretation of the second step needs a representative with a specific complete geometry.

Third, the conclusion is not without precedent in spirit. In double holography, a brane with induced gravity and a boundary CFT supply just the data our hierarchy lacks, and a bulk, a brane and a boundary do compose [13, 14]. Read this way, the selection map $\mathcal{S}$ is the geometric shadow of brane or interface data, and the “relative sector” of the abstract is not a rival to those constructions but a conformal-geometric reformulation of why they need that data.

Finally, the scalar test of Section 6 presupposes a wave equation for $\phi_{(0)}$ on (M, γ_+) that first-level holography does not provide, and the candidate principles for $\mathcal{S}$ are listed without any theorem showing that one of them works. These are open questions, not results.

Perhaps that is the more interesting answer. Holography may not be a ladder that can be climbed indefinitely, each boundary becoming the next bulk. Where two infinities meet, geometry recalls that the two descriptions are not independent: they intersect along a corner. What lives there may be neither another universe nor merely the shadow of the first, but a relative sector whose existence is written into the incidence of the two geometries. The failure of recursive holography is therefore not the end of the holographic idea, but a limit on how far its self-similarity can be pressed.

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